Weiqi Gai

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Research Interests

Building on robot motion planning and autonomous exploration, my research extends to target search and navigation driven by semantic understanding. I study spatial perception, semantic reasoning, decision-making and planning, and autonomous exploration in complex unknown environments, and further explore general embodied intelligence through vision-language-action (VLA) and world action models (WAM). My future research will focus on general-purpose robot navigation and embodied decision-making to improve task generalization in complex environments.

Education & Internships

Differential Robotics (Hangzhou) Co., Ltd. Aug. 2025 – Present
Large-Model Algorithm Intern — Intelligent UAV Navigation

Beihang University Sept. 2024 – Present
Master's Student — Aircraft Design — Intelligent UAVs Advisor: Prof. Zhijun Meng

Zhejiang University Huzhou Institute, FAST-LAB June 2024 – Sept. 2024
Research Assistant — Autonomous UAV Exploration Advisor: Prof. Fei Gao

Beihang University Sept. 2020 – June 2024
Undergraduate — Aircraft Environment Control and Life Support

Publications & Research Outputs

1. **OSET: Fully Onboard Semantic Exploration for UAVs with Traversability-Aware Scene Graph**
   Gai, W., Gao, Y., Zhou, Y., Wu, Y., Zhou, X., Zhao, G., Wang, J., Gao, F. & Meng, Z.
   *Chinese Journal of Aeronautics (**CJA; CAS Q1; IF 6.9**), 2026, **Under Review**.*
   - **Capability:** Given a natural-language instruction such as "find the sofa in the living room," a UAV autonomously prioritizes search regions, avoids obstacles, and locates the target in an unknown indoor environment. Perception, reasoning, and planning run entirely onboard.
   - **Technical Approach:** A lightweight scene graph integrates traversable-space topology, regional relations, and open-vocabulary targets. An onboard LLM selects high-value regions, and local frontier planning produces safe, executable flight paths.
   - **Results:** Relative improvements of approximately **54%** in success rate and **65%** in SPL over the baseline on long-horizon Unity tasks; fully onboard deployment on a UAV equipped with an Orin NX.

2. **NavDreamer: Video Models as Zero-Shot 3D Navigators**
   Huang, X.†, **Gai, W.**†, Wu, T., Wang, C., Zheng, Q., et al.
   *IEEE Robotics and Automation Letters (**RA-L; CAS Q2; IF 5.3**), 2026, **Accepted**. †Co-first authors (Gai listed second).*
   - **Capability:** Without retraining for new scenes or targets, a robot uses its current camera image and a natural-language instruction to avoid obstacles, find objects, and reach specified spatial locations in unfamiliar indoor and outdoor 3D environments.
   - **Technical Approach:** A generative video model predicts multiple future navigation videos. A VLM ranks them by safety, scene consistency, and task completion; an inverse dynamics model and metric depth then recover executable 3D waypoints.
   - **Results:** Approximately **85%** overall success across **20 categories of real-world indoor and outdoor tasks**; outdoor relative-scale error reduced from **54%** to **6%**.

3. **SwiftGuide: Observation-Preserving Hierarchical Multimodal Semantic Mapping for User-Aligned Embodied Navigation Agents**
   Huang, Z., **Gai, W.**, Zhou, X., Wu, Y., Gao, F. & Lyu, X.
   *Manuscript, 2026, **In Preparation**.*
   - **Capability:** Users retrieve previously observed objects by category, free-form description, reference image, or personalized name. After the user confirms or corrects a candidate, the UAV autonomously returns to the object's location.
   - **Technical Approach:** Pose-tagged raw RGB observations form a three-level Agent Hierarchical Memory (AHM). A VLM supports multimodal retrieval and user-preference memory, while an incremental Height-sampled Generalized Voronoi Diagram (HiGVD) enables safe path planning.
   - **Results:** Across **81 explorations and 965 multimodal query units**, description-query Hit@5 reached **60.8%** (best baseline: 39.0%). Navigation success across 267 reachable targets reached **84.6%**, exceeding the best baseline by 8.6 percentage points.

Publications & Research Outputs (continued)

4. **Grounding Human Interventions via Anchored Corrective-Measure Estimation**
   Cui, X., Zhu, M., **Gai, W.**, Wu, Y., Zhou, X., Gao, F., Wang, B., Zhou, W. & Li, H.
   ***AAAI (CCF-A)**, 2027, **Under Review**.*
   - **Capability:** Sparse human takeovers before collisions or operational errors teach safer actions. Ground robots and quadrotors subsequently improve autonomous obstacle avoidance and require less continuous human monitoring.
   - **Technical Approach:** Expert and original-policy actions are paired under the same observation at intervention time. Corrective weights depend on risk and action differences; context-aware optimal transport reuses reliable supervision across similar states.
   - **Results:** Across three simulation benchmarks and real-world ground-robot and quadrotor obstacle avoidance, average success improved by **34.3 percentage points** over PPL, with **25.3%** less expert supervision.

5. **Efficient Autonomous UAV Exploration Framework With Limited FOV Sensors for IoT Applications**
   Wang, Z., Meng, Z., Tian, T., **Gai, W.**, Zhao, G., et al.
   *IEEE Internet of Things Journal (**CAS Q1; IF 8.7**), 12(1): 713–725, 2025, **Accepted**.*
   - **Capability:** A UAV with a limited-field-of-view depth camera autonomously selects flight destinations and incrementally maps unknown 3D environments without prior manual mapping, supporting environmental monitoring, data collection, and search and rescue.
   - **Technical Approach:** Random-seed frontier generation and anisotropic Gaussian viewpoint sampling rapidly identify unexplored regions. A sliding-window TSP determines the global visiting order, while active-perception trajectory optimization jointly considers flight and observation.
   - **Results:** Exploration time reduced by **17%–27%** and flight-path length by **10%–17%** in simulation and real-world environments.

Project Experience

OSET: Fully Onboard Semantic Exploration for UAVs Aug. 2025 – May 2026
***Differential Robotics Technology Co., Ltd.** / Project Lead; First Author*

- Designed an integrated pipeline of **traversability-aware scene graphs, open-vocabulary semantic perception, onboard LLM-based region selection, and local exploration planning** for natural-language object search in unknown indoor environments, achieving a fully onboard perception, reasoning, planning, and control loop.
- **Independently completed** topic selection and problem definition, technical roadmap development, core implementation, hardware-software integration, UAV platform construction and debugging, experimental design and analysis, and manuscript writing, covering the entire development cycle from research prototype to real-flight validation.
- Submitted the work to Chinese Journal of Aeronautics as first author. Core modules became foundational capabilities for multiple company projects and operated reliably in project demonstrations supporting fundraising, transferring the research prototype into operational systems.

Navigation World Model: Inference Acceleration & Deployment July 2026 – Sept. 2026
***Differential Robotics Technology Co., Ltd.** / Core Developer*

- Built a three-stage **SFT → SCM/RCM Warmup → DMD** training pipeline for low-latency deployment of generative world models in UAV products: navigation-video generation, stable few-step sampling, and distribution-matching distillation.
- Reduced diffusion denoising from **30 steps to approximately 4**. System-level optimization using sparsification, distillation, and a lightweight TAE preserved navigation semantics and temporal continuity while reducing total generation time from 171.568 s to 0.702 s (**244.4× speedup**), providing a foundation for near-real-time deployment.

Qwen3.5 Post-Training & Inference Evaluation (Mathematics) Sept. 2026
Research and Learning Project

- Built **Attention LoRA, Attention+MLP LoRA, and Full SFT** post-training pipelines with **Megatron Bridge + Megatron Core** on four NVIDIA H20 GPUs to improve mathematical reasoning. Curated and deduplicated multisource mathematical data into a reproducible training set of 500,000 examples.
- Developed automated evaluation with **parallel vLLM inference services and an HTTP client**, evaluating and visualizing Direct/CoT results for each of 1,819 GSM8K/Math500 questions. Relative to the baseline, the best checkpoint raised single-pass accuracy from 41.4% to 55.1% and CoT accuracy from 36.3% to **60.0%**.

Reproducing the LeRobot Manipulation Learning Pipeline 2026
Research and Learning Project / Real-Robot Reproduction

- Collected teleoperated demonstrations on a LeRobot arm for XT60 aviation-connector grasping, creating **200 real-robot demonstrations** and completing data preprocessing, training, inference deployment, and closed-loop testing.
- Reproduced **ACT, Diffusion Policy, π_0, and $\pi_{0.5}$** in sequence, covering the manipulation pipeline from imitation learning and diffusion policies to VLA, and achieving end-to-end connector grasping from visual observations.

Project Experience (continued)

FlyAgent: General-Purpose Embodied Flight Agent Nov. 2025 – Present
Differential Robotics Technology Co., Ltd. */ Core Developer*

- As a principal contributor, developed a general-purpose embodied agent that controls UAVs through web-based dialogue, translating natural-language instructions into **navigation, manipulation, exploration, target tracking, and automated inspection** tasks.
- Designed the core system architecture, network communication topology, and onboard infrastructure. Connected the web interface, agent services, task orchestration, and flight-control execution to support reliable coordination across modules.
- Debugged agent tool calling and task capabilities, and extended CoPaw. The system has reached stable operation, with productization underway.

Large-Scale Heterogeneous Robot Swarm Exploration June 2024 – Sept. 2024
Zhejiang University Huzhou Institute, FAST-LAB */ Core Developer*

- Built a **15-robot heterogeneous swarm** comprising vision-based quadrotors, LiDAR quadrotors, coaxial twin-rotor UAVs, and radar-equipped Ackermann-steered ground vehicles for collaborative missions in multistory buildings with varied lighting, dense obstacles, narrow passages, and dust.
- Led autonomous UAV exploration in unknown environments, inter-robot map exchange, and flight-platform hardware debugging, integrating exploration planning, shared communication, and real-robot operation.
- Replaced FUEL's global occupancy-grid exploration with **local grids, a global large-scale auxiliary grid, and topological global paths**, reducing dependence on global dense maps during large-scale exploration.

Large-Scale Autonomous Obstacle Avoidance for an Unmanned Helicopter Dec. 2024 – June 2025
AVIC Institute 602 */ Project Lead*

- Adapted algorithms for autonomous obstacle avoidance and engineering deployment of an unmanned helicopter in large-scale environments, and validated equivalent functionality on a multirotor platform.
- Extensively customized a multirotor UAV demonstrator, modified obstacle-avoidance algorithms, and developed and deployed platform-specific engineering code, connecting perception, planning, control, and integrated vehicle testing.
- Led requirements coordination, task decomposition, scheduling, and core development; delivered on schedule and successfully completed the project.

Honors & Awards

- Outstanding Graduate Student, Beihang University, 2026.
- Second-Class Master's Scholarship, Beihang University, 2026.
- First-Class Master's Scholarship, Beihang University, 2025.
- Meritorious Winner (M Award), Mathematical Contest in Modeling / Interdisciplinary Contest in Modeling (MCM/ICM).

Personal Strengths & Skills Overview

- **Research Experience:** A solid foundation in mobile-robot motion planning, with sustained research in UAV motion planning and autonomous exploration from my undergraduate thesis through master's research.
- **Engineering Implementation:** Focused on deploying algorithms in practice, with **full-stack real-robot development and debugging experience** spanning UAV 3D structural design, PCB design, flight-controller tuning, system integration and deployment, and engineering performance optimization.
- **Project Collaboration:** Extensive industry internship and cross-team experience, including project coordination, task decomposition, schedule management, and team management.
- **Rapid Learning:** Strong learning and research skills; able to quickly follow emerging research, reproduce key techniques, and develop engineering solutions, with sustained curiosity about new knowledge.

Robot Planning: Progressing from robot motion planning to VLA and WAM to advance general-purpose robot navigation and decision-making.
Embodied AI: Familiar with LLM/VLM fundamentals and post-training methods, as well as world-model architectures and model distillation.
Engineering: Proficient in C++, Python, ROS, and efficient AI Harness workflows; skilled in SolidWorks, Altium Designer, and MATLAB.
Collaboration: Experienced as a project lead and core developer, with skills in task decomposition, schedule management, and cross-team collaboration.